

# NOTES 14: RANDOMIZATION TESTS AND P-VALUES

Stat 120 | Spring 2026

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**Example:** Resume callbacks. Researchers want to know if unconscious gender bias affects hiring decisions. They created 20 identical, highly qualified resumes. On 10 of the resumes, they put a traditionally male name (like “John”). On the other 10, they put a traditionally female name (like “Emily”). They submitted all 20 resumes to open job postings and waited to see who received a callback for an interview. Of the 20 resumes, 10 received callbacks. 7 of those callbacks went to “John” and 3 went to “Emily”. Is this evidence of gender bias?

- Population parameter:
- Sample statistic:
- Hypotheses:

How do we gauge whether we have strong evidence against the null hypothesis, or if the difference could be due to sampling variability?

Procedure:

- 10 cards labeled “Yes” (those that received a callback)
  - 10 cards labeled “No” (did not receive a callback)
1. Verify you have 10 yes’s and 10 no’s
  2. Shuffle the 20 cards together thoroughly
  3. Deal the cards face down into two equal piles of 10 (Pile 1 is the “Emily” group, Pile 2 is the “John” group)
  4. Flip over the “John” group and count the number of yes’s
  5. Record your result and then pass the cards to the next person

Your group should perform the shuffle and deal 4 times (1 per person). Record the number of “Yes” cards in the Male Name pile for each trial.

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| Trial Number | Number of Callbacks (“Yes” Cards) in the “John” Pile | Sample Statistic |
|--------------|------------------------------------------------------|------------------|
| Trial 1      |                                                      |                  |
| Trial 2      |                                                      |                  |
| Trial 3      |                                                      |                  |
| Trial 4      |                                                      |                  |

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## **StatKey Example:** Resume callbacks

### **Constructing null distribution for a difference in proportions**

Combine all cases, then randomly assign each case to one of two groups with the same sample sizes as the original data. Record the difference in proportions between the two groups. Do this many times.

**StatKey Example** In a random sample of 765 adults in the United States, 322 say they could not cover a \$400 unexpected expense without borrowing money or going into debt. A journalist claims that the true proportion is actually 50%. Is the journalist justified?

### **Constructing null distribution for a single proportion**

In this example, flip a coin 765 times and record the proportion that came up heads. Do this many times to construct the randomization distribution

**R Example** In a study about Universal Basic Income, the SEED institute randomly selected 100 residents and gave them an unrestricted \$500 a month for 24 months (the “UBI” group). At the end of the study period, they were given an emotional well-being assessment and results were compared to 100 control-group residents who did not receive payments. The average well-being score among the UBI group was 31.5 and among the control group was 29.5. Does UBI result in higher well-being scores?

```
ubi |>
  group_by(group) |>
  summarize(
    mean = mean(well_being),
    n = n()
  )
```

```
# A tibble: 2 x 3
  group    mean     n
  <chr>   <dbl> <int>
1 Control 29.5   100
2 UBI     31.5   100
```

```
library(CarletonStats)
permTest(well_being ~ group, data = ubi, alternative = "greater", seed = 101525)
```

**\*\* Permutation test \*\***

Permutation test with alternative: greater

Observed statistic

Control : 29.54 UBI : 31.49

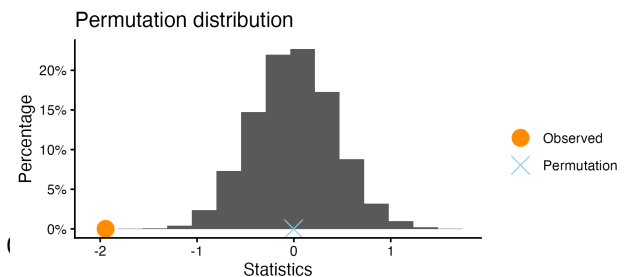
Observed difference: -1.944

Mean of permutation distribution: -0.00282

Standard error of permutation distribution: 0.9999999

P-value: 1

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### Constructing null distribution for a difference means

Combine all cases, then randomly assign each case to one of two groups with the same sample sizes as the original data. Record the difference in means between the two groups. Do this many times.

**Be careful about direction of hypotheses!**

## Statistical Significance

### Significance Level

Cutoff value for rejecting  $H_0$  (called “alpha” or  $\alpha$ )

If p-value \_\_\_\_\_  $\alpha$ , we “reject  $H_0$ ” and say results are “statistically significant”

Common values:

Default:

### Statistical Significance

If p-value is small enough, it’s very unlikely to get the results we observed if  $H_0$  is true due to sampling variability

Results are statistically significant if they provide convincing evidence against  $H_0$

### Formal Statistical Decisions

- p-value \_\_\_\_\_  $\alpha$ : Reject  $H_0$
- p-value \_\_\_\_\_  $\alpha$ : Do not reject  $H_0$

## Summary

1. Formulate hypotheses in terms of *population parameter*
2. Collect data and compute a *sample statistic*
3. Use the *sample statistic* to make a claim about the *hypotheses*

Today, we covered step 3:

3a. Construct *null/randomization distribution* using permTest in R or StatKey

If  $H_0$  is \_\_\_\_\_, what does the range of sample statistics look like?

3b. Compute *p-value*

Numeric summary of how extreme \_\_\_\_\_  
is compared to \_\_\_\_\_ distribution

3c. Make a formal decision about  $H_0$

- Reject  $H_0$  if \_\_\_\_\_
- Do not Reject  $H_0$  \_\_\_\_\_

3d. Report your conclusion in context